Day:

Date:

Q1. Number System Conversions

Q Convert $(756.34)_8$ to Decimal

A $= 7\times8^2+5\times8^1+6\times8^0+3\times8^{-1}+4\times8^{-2}$

$= 7\times64+5\times8+6\times1+0.375+0.0625$

$= 448+40+6+0.375+0.0625$

$= (756.34)_8 = (494.4375)_{10}$

Q Convert $(245)_{10}$ to binary and hexadecimal

A To Binary:-

2	245
2	122-1
2	61-0
2	30-1
2	15-0
2	7-1
2	3-1
	1-1

$= (11110101)_2$

Date: Day:

To Hexadecimal:-

16	245
	15-5

$(F5)_{16}$ ∵ (15=F)

Q Convert $(101101.101)_2$ to Decimal

A $= [1\times2^5 + 0\times2^4 + 1\times2^3 + 1\times2^2 + 0\times2^1 + 1\times2^0 + 1\times2^0 + 0\times2^{-1} + 0\times2^{-2}]$

$= 1\times2^5 + 0\times2^4 + 1\times2^3 + 1\times2^2 + 0\times2^1 + 1\times2^0 + 1\times2^{-1} + 0\times2^{-2} + 1\times2^{-3}$

$= 32 + 8 + 4 + 1 + 0.5 + 0.125$

$= (45.625)_{10}$

Q Convert $(3A7)_{16}$ to Binary and Decimal

To Binary:-

3 = 0011

A = 1010

7 = 0111

So, $(3A7)_{16} = (001110100111)_2$

To Decimal :-

$= (3 \times 16^2) + (10 \times 16^1) + (7 \times 16^0)$

$= (3 \times 256) + (10 \times 16) + (7 \times 1)$

$= 786 + 160 + 7$

$= (935)_{10}$

Q2. Find 1s and 2s Complements

Q 10101100

A 1s Complement: 01010011

2s Complement: 01010011 (1 1)

+ 1

01010100

Q 01101010

A 1s Complement: 10010101

2s Complement: 10010101 (1)

+ 1

10010110

Date: Day:

Represent the following numbers in 8-bit 2's complement form.

Q -45

A +45 in binary = ~~0010 0100010101~~

0010 1101

1's Complement: 1101 0010

2's Complement: 1101 0010

1

1101 0011

Q -92

A +92 in Binary = 0101 1100

1's Complement = 1010 0011

2's Complement = 1010 0011

1

10100100

Q +78

+78 in Binary = 0100 1110
Positive so this is the answer.

Subtract using 2's Complement method.

Q 34-19

A 34 in bits = 0010 0010
+19 in bits = 0001 0011
1's complement of 19: 1110 1100
2's Complement of 19: 1110 1100
+ 1
1110 1101

34-19 = 0010 0010
+ 1110 1101
10000 1111
In, 8-bit System, discard the 9th bit
= 0000 1111

Q 19-34

A +19 in bits = 0001 0011
+34 in bits = 0010 0010
1's Complement of 34: 1101 1101

Date:

2's Complement of 34: 1101 1101

+ 1

1101 1110

19-34 = 0001 0011

1101 1110

1111 0001

Q -25-17

A 25 in bits: 0001 1001

1's Complement of 25: 1110 0110

2's Complement of 25: 1110 0110

+1

1110 0111

17 in bits: 0001 0001

1's Complement of 17: 1110 1110

2's Complement of 17: 1110 1110

+1

1110 1111

-25-17 = 1110 0111

1110 1111

1 1101 0110

Dropping 9th bit = 1101 0110

Date: Day:

Q 45 - (-18)

A $45 = (0010\ 1101)_2$

18 in bits = 0001 0010

1's Complement of 18: 1110 1101

2's Complement of 18: 1110 1101

1

11101110

45 - (-18) = 45 + 18

= 0010 1101

+ 0001 0010

0011 1111

Q3 Binary Fraction Conversion

Q $(110101.1011)_2$ to Decimal

A $= 1x2^5 + 1x2^4 + 0x2^3 + 1x2^2 + 0x2^1 + 1x2^0 + 1x2^{-1} + 0x2^{-2} + 1x2^{-3} + 1x2^{-4}$

$= 32 + 16 + 0 + 4 + 0 + 1 + 0.5 + 0 + 0.125 + 0.0625$

$= 53 + 0.6875$

$= (53.6875)_{10}$

Q Convert $(0.375)_{10}$ to binary and verify by converting back to decimal

A $0.375 \times 2 = 0.75 \rightarrow 0$

$0.75 \times 2 = 1.5 \rightarrow 1$

$0.5 \times 2 = 1 \rightarrow 1$

$= 0.011$

Verifying :-

$= 0 \times 2^{0} + 0 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3}$

$= 0 + 0 + 0.25 + 0.125$

$= (0.375)_{10}$

Q Convert $(1011.011)_2$ to Octal

A $(001\ 011.\ 011)_2$

$001 = 1$, $011 = 3$

$= (13.3)_8$

Q4 Perform: Octal Addition (Base 8)

Q $(56)_8 + (27)_8$

A
$$\begin{array}{r} {\scriptstyle 1} \\ 56 \\ +27 \\ \hline 105 \end{array}$$

$= (105)_8$

Q $(135)_8 + (247)_8$

A
$$\begin{array}{r} {\scriptstyle 1}\,{\scriptstyle 1} \\ 135 \\ +247 \\ \hline 404 \end{array}$$

$= (404)_8$

Q $(672)_8 + (354)_8$

A
$$\begin{array}{r} {\scriptstyle 1} \\ 672 \\ +354 \\ \hline 1246 \end{array}$$

$= (1246)_8$

Q $(507)_8 + (468)_8$

A $\overset{①}{5}07$

$+468$

1177

$= (1177)_8$

Q $(777)_8 + (1)_8$

A $\overset{①}{\overset{①}{7}}77$

$+ \quad 1$

1000

$= (1000)_8$

Q $(645)_8 + (732)_8$

A 645

$+732$

1577

$= (1577)_8$

Date: Day:

Q Perform Hexadecimal Addition (Base 16)
(A=10, B=11, C=12, D=13, E=14, F=15)

Q $(A7)_{16} + (3D)_{16}$

A
$$\begin{array}{r} {}^{1} \\ A\ 7 \\ +\ 3\ D \\ \hline E\ 4 \end{array}$$
$= (E4)_{16}$

Q $(5F)_{16} + (8C)_{16}$

A
$$\begin{array}{r} {}^{1} \\ 5F \\ +\ 8C \\ \hline E\ B \end{array}$$
$= (EB)_{16}$

Q $(9B)_{16} + (7E)_{16}$

A
$$\begin{array}{r} {}^{1} \\ 9B \\ +\ 7E \\ \hline 119 \end{array}$$
$= (119)_{16}$

Date: Day:

Q $(2AF)_{16} + (19C)_{16}$

A
$$\begin{array}{r} {\scriptstyle 1\ 1\ \ } \\ 2AF \\ +19C \\ \hline 44B \end{array}$$

$= (44B)_{16}$

Q $(FFF)_{16} + (1)_{16}$

A
$$\begin{array}{r} {\scriptstyle 1\ 1\ \ } \\ FFF \\ +\ \ 1 \\ \hline 1000 \end{array}$$

$= (1000)_{16}$

Q $(ABC)_{16} + (DEF)_{16}$

A
$$\begin{array}{r} {\scriptstyle 1\ 1\ \ } \\ ABC \\ +DEF \\ \hline 18AB \end{array}$$

$= (18AB)_{16}$

Octal ~~Add~~ Subtraction

Q $(65)_8 - (27)_8$

A
$$\begin{array}{r} {}^{5}6\overset{8}{5} \\ 2\,7 \\ \hline 3\,6 \end{array}$$

$= (36)_8$

Q $(154)_8 - (67)_8$

A
$$\begin{array}{r} {}^{4}1\overset{8}{5}\overset{6}{4} \\ 6\,7 \\ \hline 6\,5 \end{array}$$

$= (65)_8$

Q $(703)_8 - (256)_8$

A
$$\begin{array}{r} {}^{6}{}^{1}7\overset{8}{0}\overset{8}{3} \\ 2\,5\,6 \\ \hline 4\,2\,5 \end{array}$$

$= (425)_8$

Date: _____ Day: _____

Q $(1000)_8 - (1)_8$

A
$$\begin{array}{r} 1000 \\ -\quad 1 \\ \hline 0777 \end{array}$$

$= (777)_8$

Q $(742)_8 - (356)_8$

A
$$\begin{array}{r} 742 \\ 356 \\ \hline 364 \end{array}$$

$= (364)_8$

Q: $(742)_8 - (365)_8$

A:
$$\begin{array}{r} 742 \\ 365 \\ \hline 355 \end{array}$$

$= (355)_8$

Q $(600)_8 - (457)_8$

A
$$\begin{array}{r} 600 \\ 457 \\ \hline 121 \end{array}$$

$= (121)_8$

Date:

Hexadecimal Subtraction

Q $(A5)_{16} - (3C)_{16}$

A
$$\begin{array}{r} {}^{9}\,{}^{(16)} \\ A\,5 \\ -\,3\,C \\ \hline 6\,9 \end{array}$$

$= (69)_{16}$

Q $(7F)_{16} - (29)_{16}$

A
$$\begin{array}{r} 7\,F \\ -\,2\,9 \\ \hline 5\,6 \end{array}$$

$= (56)_{16}$

Q $(1B4)_{16} - (9D)_{16}$

A
$$\begin{array}{r} {}^{A}\,{}^{(11)} \\ 1\,B\,4 \\ -\,9\,D \\ \hline 1\,1\,7 \end{array}$$

$= (117)_{16}$

Date: Day:

Q $(200)_{16} - (1F)_{16}$

A
$$\begin{array}{r} {}^{1}{}^{15}\,2\,0\,0 \\ -\;1\,F \\ \hline 1\,E\,1 \end{array}$$

$= (1E1)_{16}$

Q $(ABC)_{16} - (5E7)_{16}$

A
$$\begin{array}{r} {}^{9}\,A\,B\,C \\ -\;5\,E\,7 \\ \hline 4\,D\,5 \end{array}$$

$= (4D5)_{16}$

Q $(1000)_{16} - (1)_{16}$

A
$$\begin{array}{r} {}^{15}{}^{15}\,1\,0\,0\,0 \\ -\;\;\;\;\;1 \\ \hline F\,F\,F \end{array}$$

$= (FFF)_{16}$

Q5 Binary Subtraction

Q 1000 - 0011

A
```
  0 0 0
  1000
- 0011
  0101
```

Q 10100 - 00101

A
```
  0 0 0 0
  10100
- 00101
  01111
```

Q 11010 - 01011

A
```
  0 0 0 0
  11010
- 01011
  01111
```

Q 10000 - 01101

A
```
  1 1 1 1
  10000
- 01101
  00011
```
= (00011)

Q 11101 - 10111

A
```
    1 1
  11101
- 10111
  00110
```
= 00110